

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

2nd Semester of B. Pharm. Examination

University Theory Examination November 2016

PH121 Pharmaceutical Biology - II

Date: 08.11.16 , Tuesday

Time: 10.00 a.m. to 01.00 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

Q 1 Attempt all questions. Each question is of one mark. 20

1. Cosmid is _____.
[A] Plasmid+Bacteriophage
[B] Plamid+HAC
[C] Plasmid+Cos-phage
[D] Plasmid+YAC
2. Bacteriophage is a/an _____.
[A] Virus
[B] Bacteria
[C] Fungi
[D] *E-Coli*
3. Plasmid plays role as a _____ in a r-DNA technology
[A] Donor DNA
[B] Cloning Vector
[C] Inducer
[D] Inhibitor
4. Lymphadenopathy is
[A] red lymph node

- [B] swollen, painful node
[C] swollen tonsils
[D] dysfunction of lymphatic system
5. Gonorrhea is a _____ type disease
[A] communicable disease
[B] non communicable disease
[C] opportunistic disease
[D] none of the above
6. Communicable disease can spread through
[A] Sneezing
[B] Sputum
[C] Both A and B
[D] Eye contact
7. Which of the following disease can spread through lymphatic system
[A] Diabetes
[B] Conjunctivitis
[C] Hemorrhage
[D] Cancer
8. Organs that filter lymph
[A] Lymph Capillaries
[B] Spleen
[C] Tonsils
[D] Lymph nodes
9. Lymph flows at
[A] low pressure and speed
[B] high pressure
[C] high speed
[D] high pressure and speed
10. The two strands in a DNA double is joined by
[A] Hydrogen bond
[B] Covalent bond

- [C] ionic bond
[D] phosphodiester bond
11. Nosocomial infections are
[A] home generated
[B] hospital generated
[C] school generated
[D] All of the above
12. A nucleotide is composed of
[A] a base+ a sugar
[B] a base+ a sugar+ phosphate
[C] a base+ a phosphate
[D] none of these
13. Tuberculosis is _____ disease
[A] Auto immune
[B] Non communicable
[C] Vector borne
[D] none of these
14. The sugar in RNA is
[A] Deoxyribose
[B] Hexose
[C] Ribose
[D] Fructose
15. Different types of pathogens responsible for communicable disease include
[A] viruses
[B] bacteria
[C] parasites
[D] all of the above
16. Lymphatic Cells include
[A] NK cells
[B] dendritic cells
[C] RBC

- Q 2 Attempt any TWO of the following
- A State and explain the Laws of Mendelian genetics 05
- B Enlist the types of DNA mutations. Explain any one in detail 05
- C Write a note on DNA replication process. 05

SECTION – II

20. Adjacent nucleotides are joined by
- [A] covalent bond
- [B] phosphodiester bond
- [C] ionic bond
- [D] peptide bond
19. Xeroderma pigmentosum results from
- [A] defect in base excision repair of damaged DNA
- [B] defect in nucleotide excision repair of damaged DNA
- [C] defect in mismatch repair of damaged DNA
- [D] defect in lac operon
18. Chromatin is composed of
- [A] nucleic acids and protein
- [B] nucleic acids only
- [C] proteins only
- [D] none of these
17. Amino acids are attached to the
- [A] acceptor arm of tRNA
- [B] anticodon arm of tRNA
- [C] codon arm of tRNA
- [D] none of these

[D] Both A and B

- Q 3 Attempt any TWO of the following**
- A Write a detailed note on Vitamin B. 05**
- B Differentiate Transcription and Translation process. 05**
- C Explain the application of r-DNA technology in the production of therapeutic agents. 05**

SECTION – III

- Q 4 Attempt any FOUR of the following**
- A Explain a composition of lymph 05**
- B Define endemic, epidemic and pandemic disease 05**
- C Enlist water distribution in human body 05**
- D Enlist Immune cells and give a detail note on it. 05**
- E Discuss any two disease condition of lymphatic system 05**
- F Explain physiology of water balance 05**

- Q 5 Attempt any FOUR of the following**
- A Basic physiology and functions of spleen 05**
- B Explain any one communicable disease in detail 05**
- C Draw a well labeled diagramme of lymph node 05**
- D Explain intracellular and extracellular ion balance 05**
- E Differentiate acquired and natural immunity 05**
- F Discuss lymph circulation 05**
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